Tree-HyperLISTA

Ultra-Lightweight Deep Unfolding for Tree-Sparse Recovery

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Why Tree-Sparse Recovery?

- Sparse recovery: reconstruct x from $y = Ax + \varepsilon$
- Tree-structured sparsity: wavelet decompositions, hierarchical representations
- Key constraint: child active $\Rightarrow$ parent must be active (tree consistency)

Gap: Classical methods (ISTA/FISTA) have no learning. LISTA (1.56M params) ignores tree structure. HyperLISTA (3 params) is elementwise only. No method combines tree structure with lightweight deep unfolding.

Tree-HyperLISTA: Method Overview

Only **3 learned hyperparameters** (c_1, c_2, c_3)

Each unfolded layer performs:

1. **Momentum:** $z = x + \beta(x - x_{prev})$
2. **Gradient:** $u = z + W^T(y - Az)$
3. **Tree scoring:** $s_i = |u_i| + \rho \Sigma s_{child}$ (bottom-up, O(n))
4. **Support select:** $S = \text{TopKTree}(s, K)$
5. **Tree threshold:** soft-threshold within S, zero outside

Three modes: M1 (hard top-K) | M2 (threshold+closure) | M3 (hybrid, recommended)

Per-layer β, θ, K derived from (c_1,c_2,c_3) via adaptive formulas — NOT learned per-layer

Key Innovation: Tree-Aware Proximal Operator

Standard (LISTA/HyperLISTA): $x_i = \text{sign}(u_i) \max(|u_i| - \theta, 0)$ → treats each coefficient independently

Tree-HyperLISTA: Score nodes via subtree info → select tree-consistent support S → threshold within S

- Same 3 global hyperparameters as HyperLISTA
- W matrix precomputed (not learned); tree topology fixed; ρ=0.5 fixed
- The ONLY change: replacing soft-threshold with tree-aware operator

→ **Structure in the operator, not in parameters**

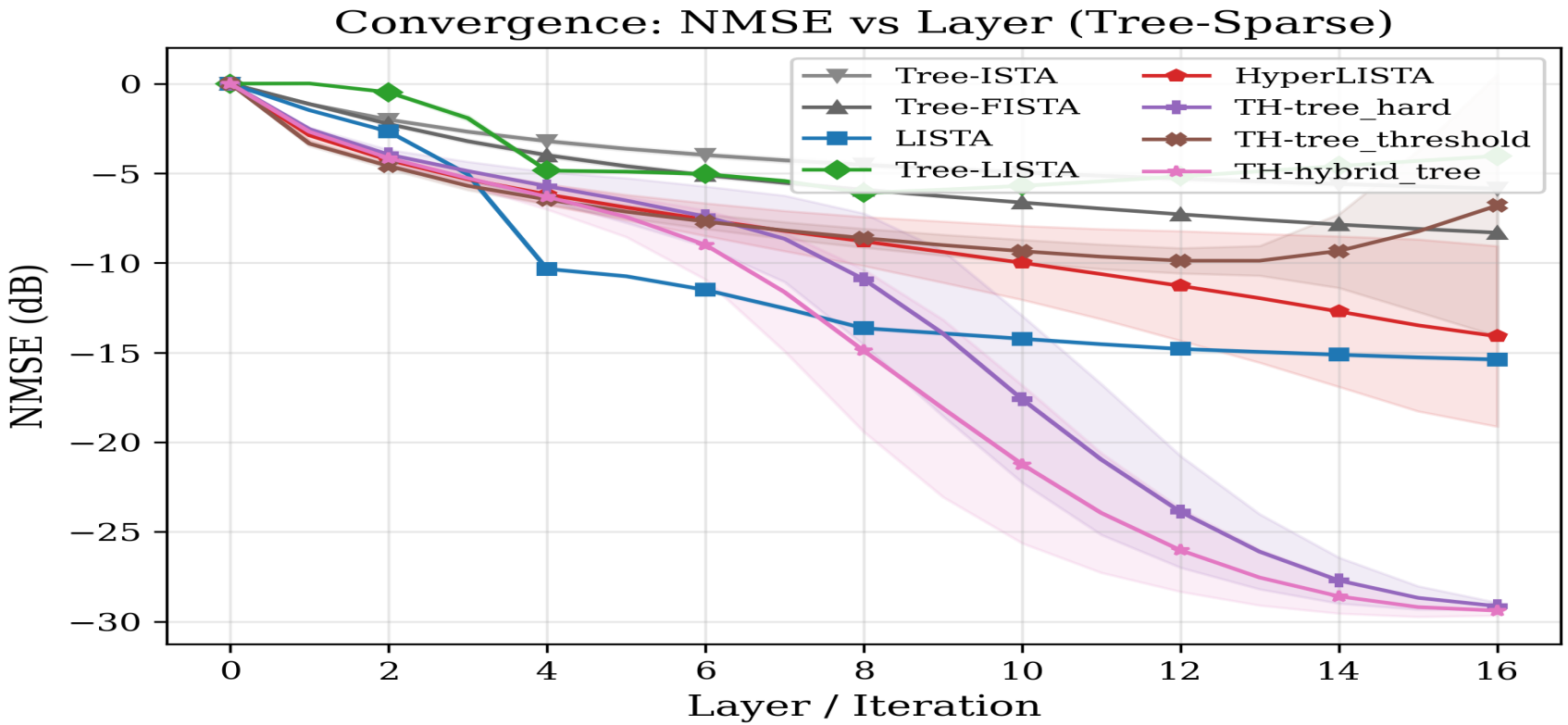
Core Results: Synthetic Tree-Sparse Recovery

n=255, m=127, K=16 layers, 30 dB SNR, 5 seeds

Model	NMSE (dB)	Precision	Recall	#Params
Tree-ISTA	–6.00 ± 0.02	0.759	0.734	0
Tree-FISTA	–8.34 ± 0.02	0.840	0.816	0
LISTA	–13.18 ± 0.08	0.255	0.997	1.56M
HyperLISTA	–15.91 ± 6.91	0.791	0.967	3
TH-Hard (M1)	**–28.80 ± 0.11**	**0.740**	**1.000**	**3**
TH-Hybrid (M3)	**–28.66 ± 0.10**	**0.663**	**1.000**	**3**

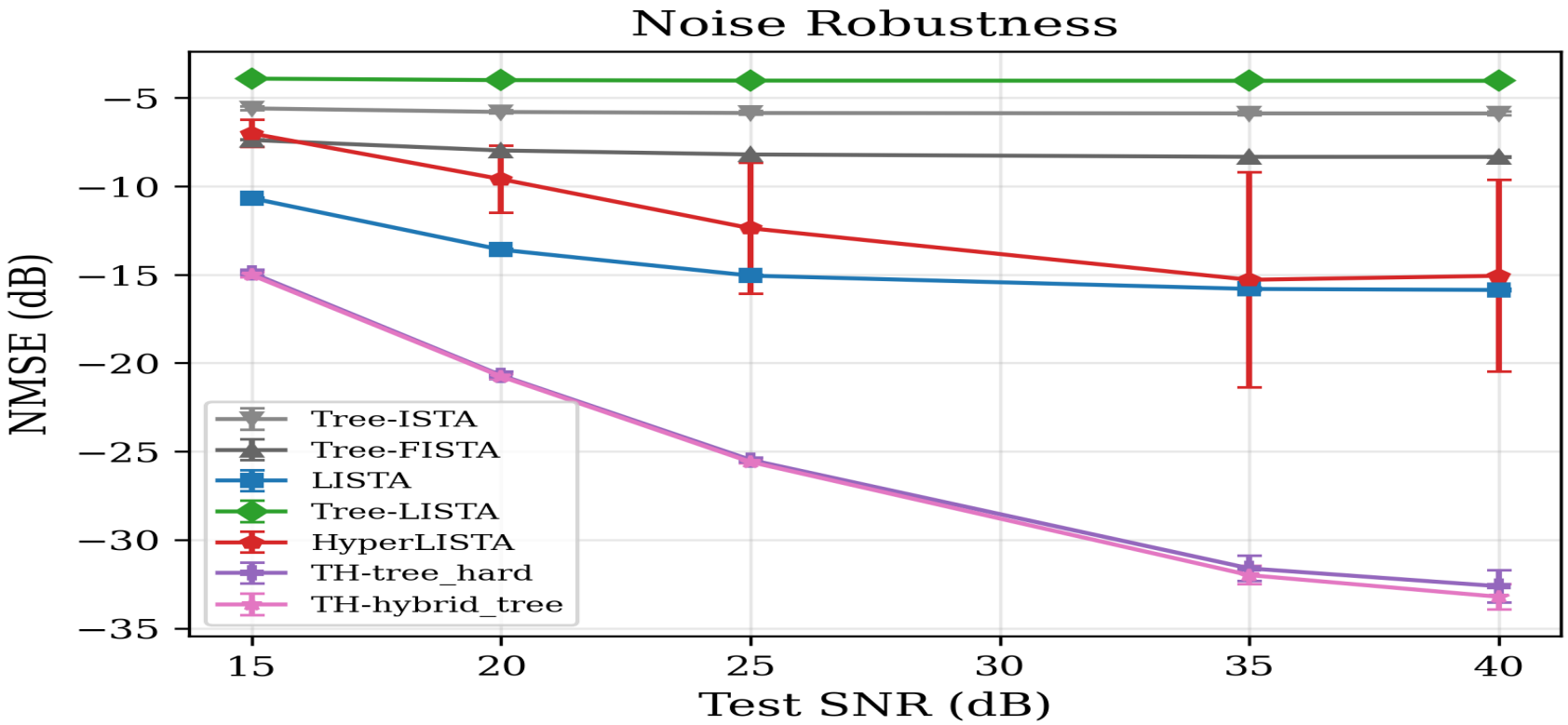
TH-Hybrid: +12.8 dB over HyperLISTA (same 3 params) | +15.5 dB over LISTA (1.56M params) | Perfect recall

Convergence & Support Recovery



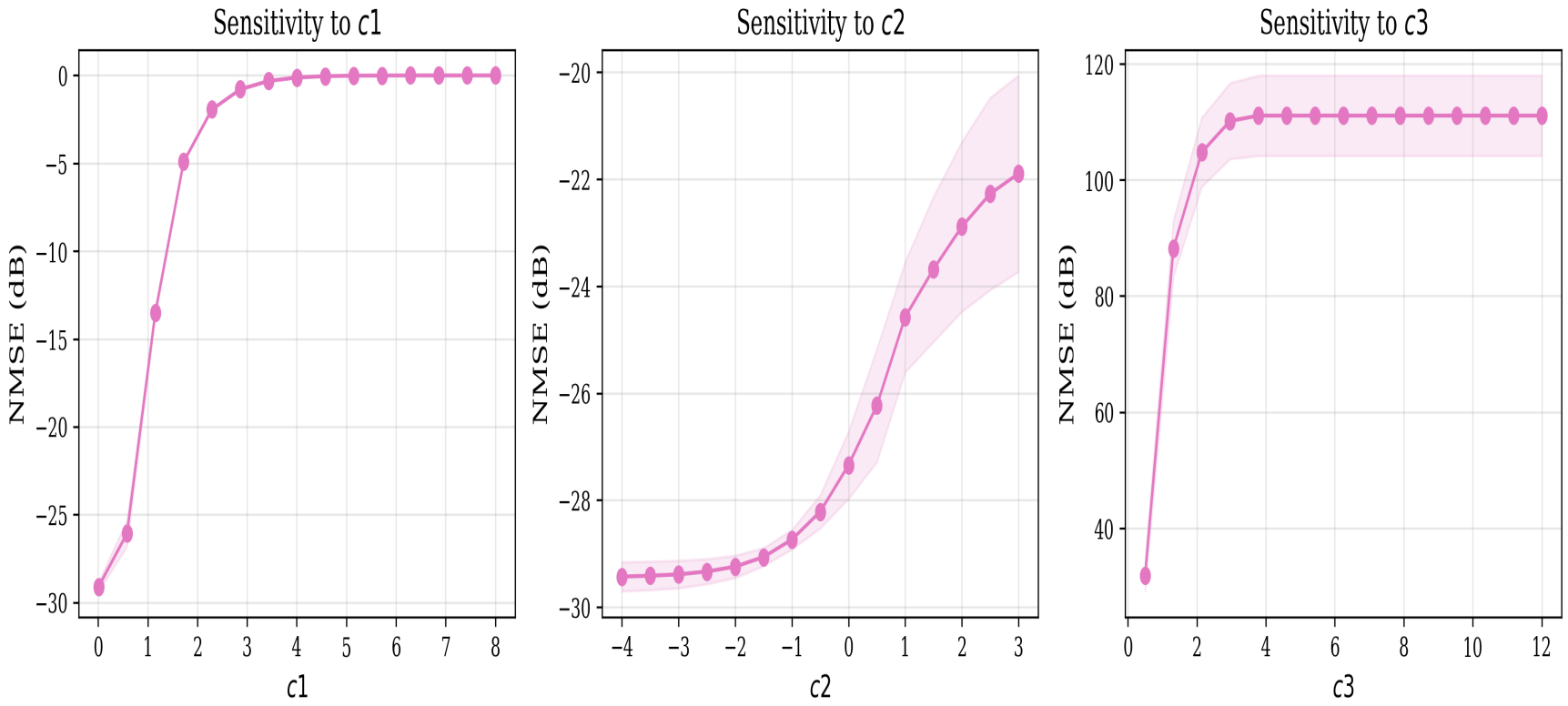
Tree-HyperLISTA converges to –28.66 dB in 16 layers with perfect support recall

Robustness: Mismatch Experiments



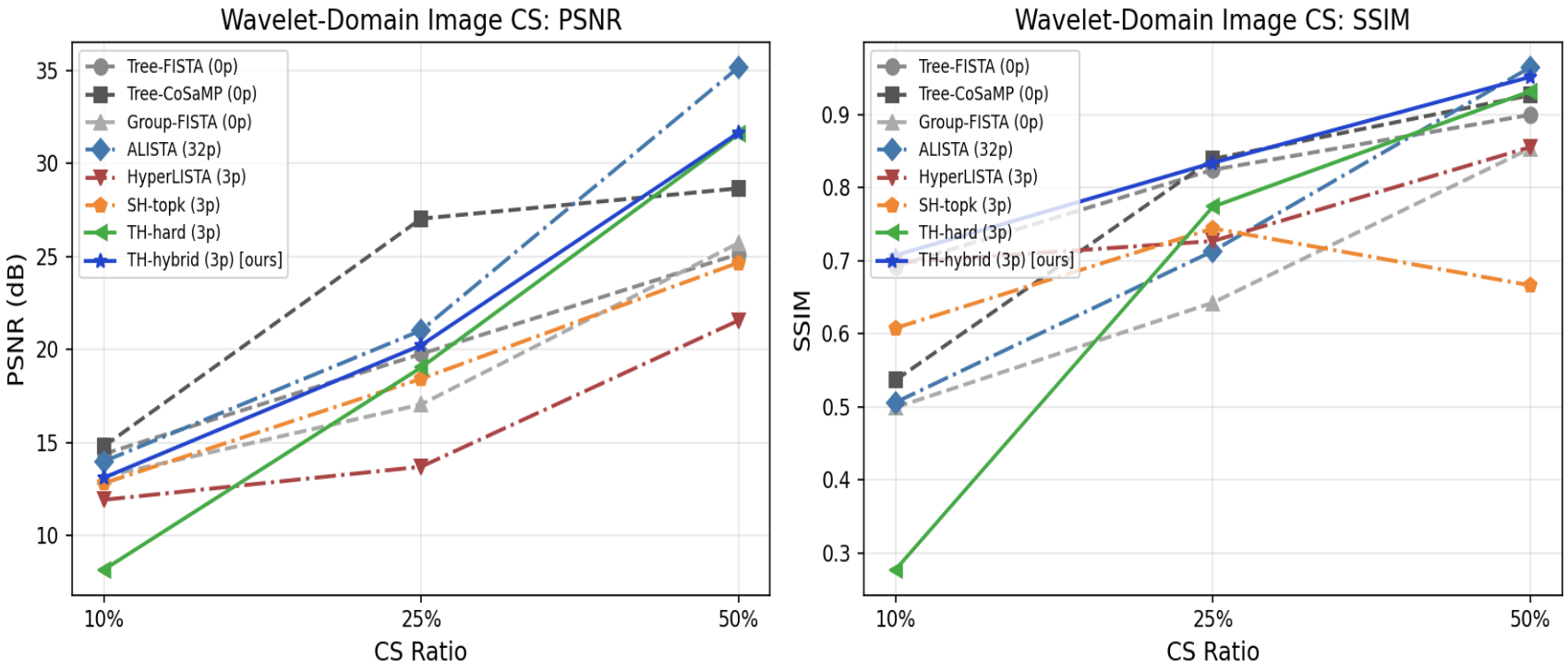
Tree-HyperLISTA degrades gracefully under SNR variation and sparsity mismatch

Ablation: Sensitivity & Mechanisms



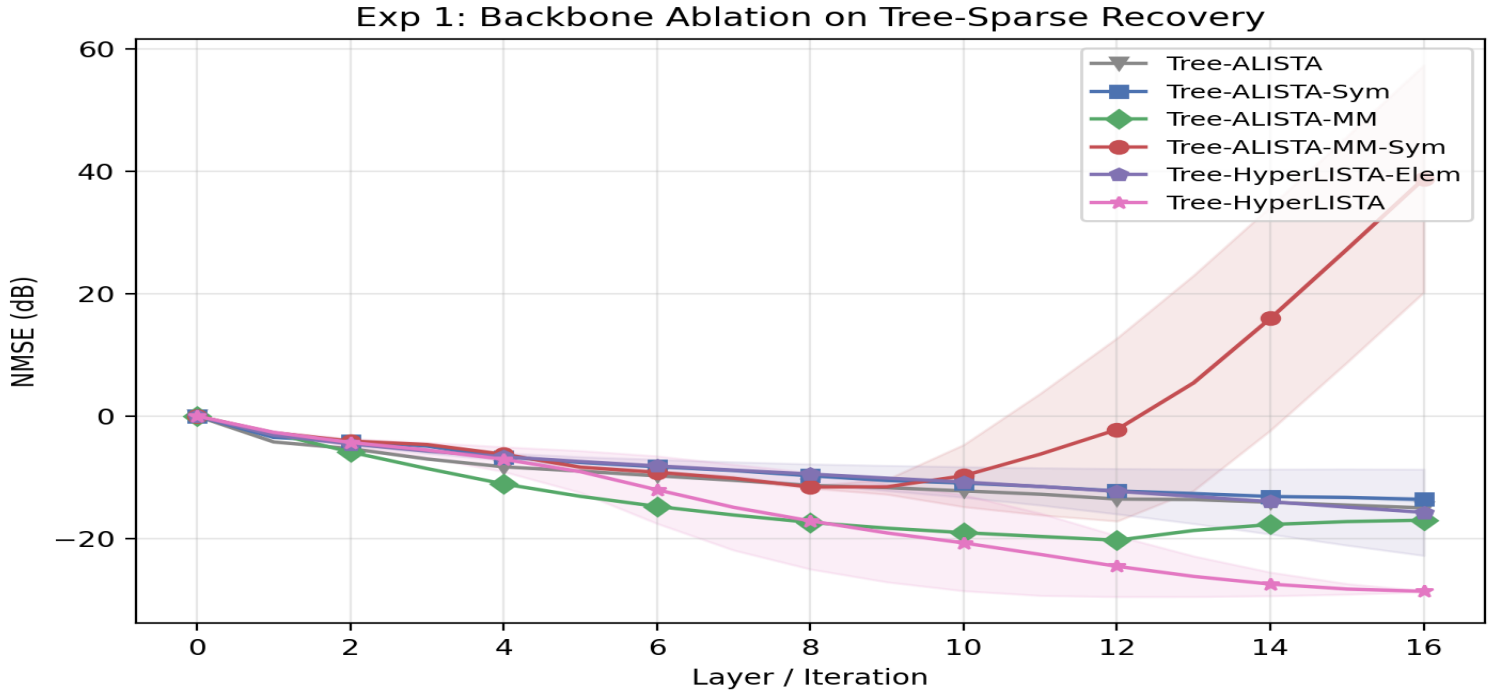
Broad optimal basin for c2 (momentum); c1 (threshold) most critical. M1 and M3 comparable.

Wavelet-Domain Image Compressed Sensing



- TH-hybrid: **best SSIM at all CS ratios** (0.707, 0.834, 0.952)
- 31.66 dB at 50% CS, beating HyperLISTA by 10+ dB with same 3 params

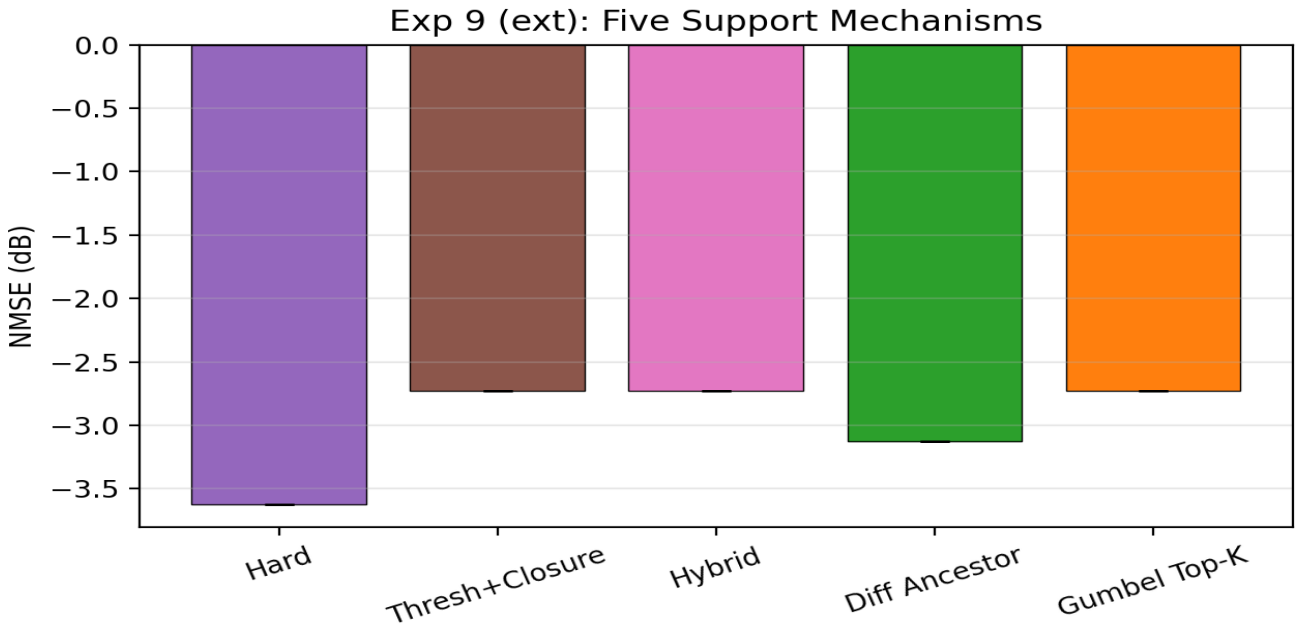
Appendix B: Backbone Ablation



Tree-ALISTA (32p): –15.00 dB | TH-Elem (3p, no tree): –15.75 dB | **Tree-HyperLISTA (3p): –28.66 dB**

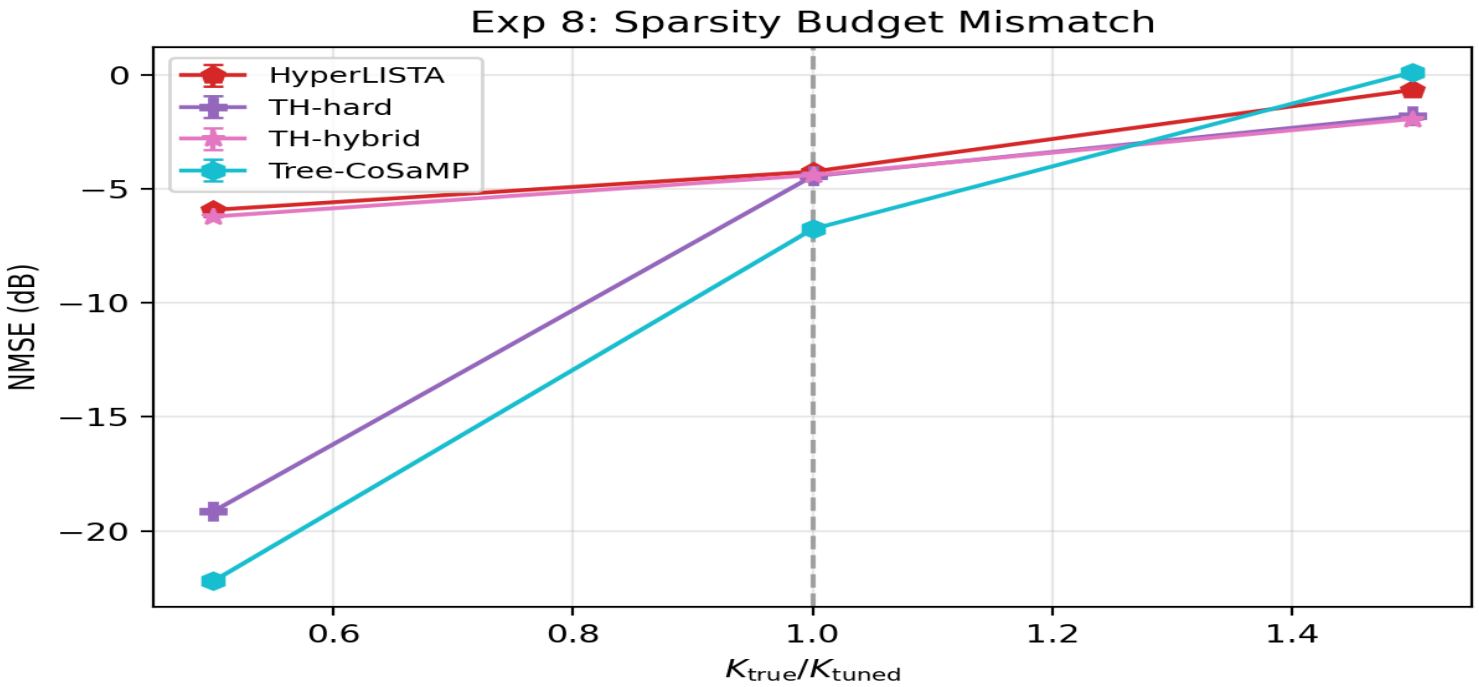
Tree-aware operator is the critical ingredient, not backbone complexity

Appendix C: Extended Mechanism Ablation



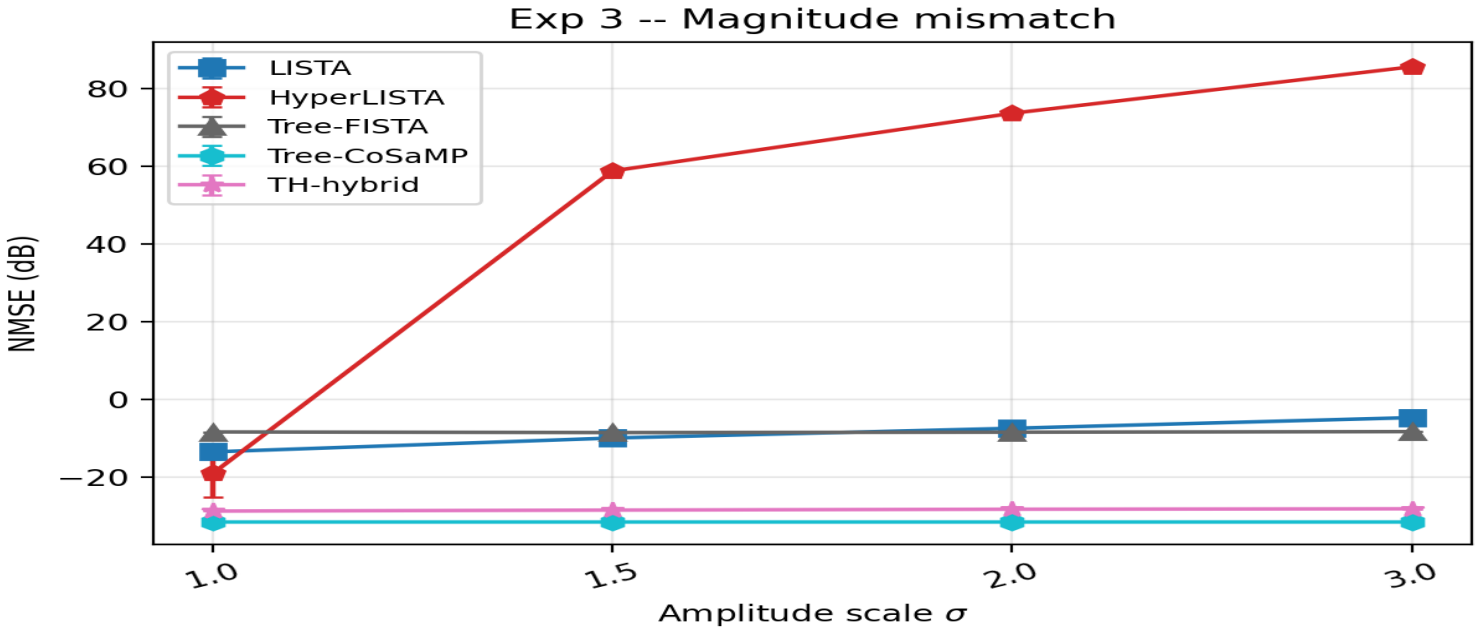
Differentiable relaxations (Diff. Ancestor, Gumbel Top-K) underperform non-differentiable hard projection

Appendix D–E: Rho Sensitivity & Sparsity Mismatch



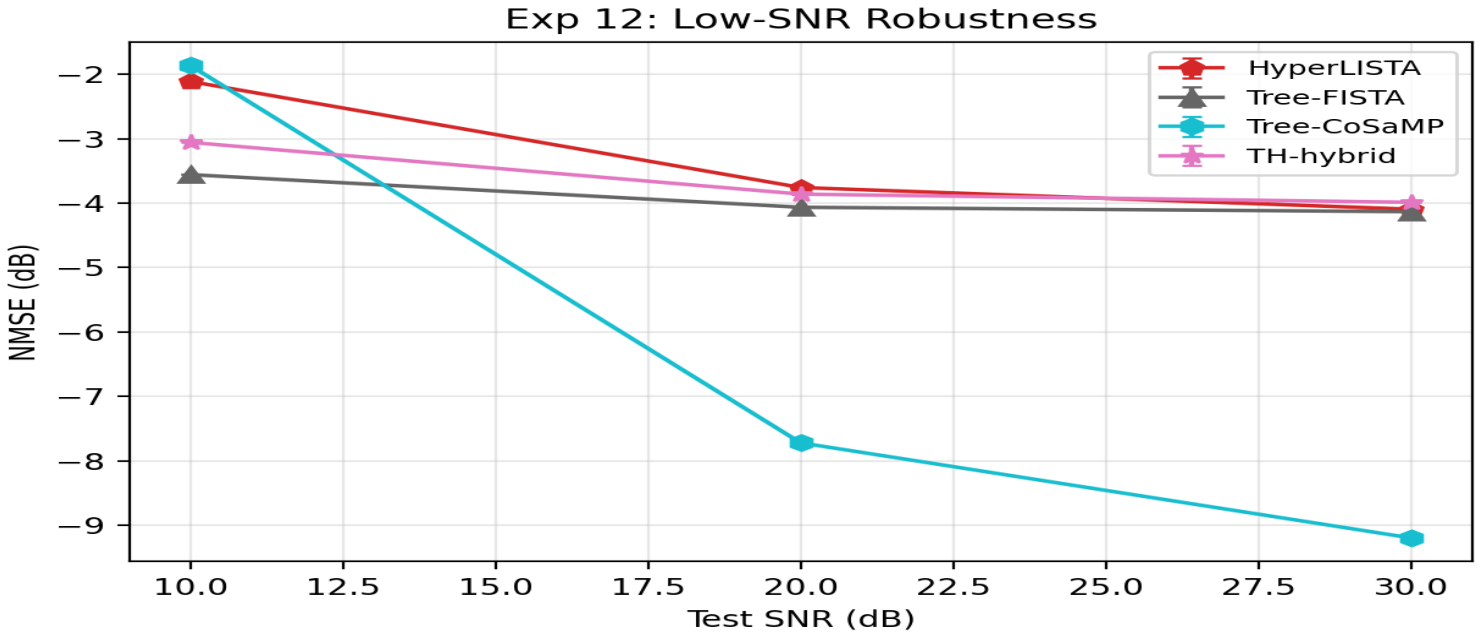
ρ=0.1 and ρ=0.5 give same result — robust to decay choice. Graceful at sparser signals; all degrade when denser.

Appendix E: Extended Mismatch Robustness



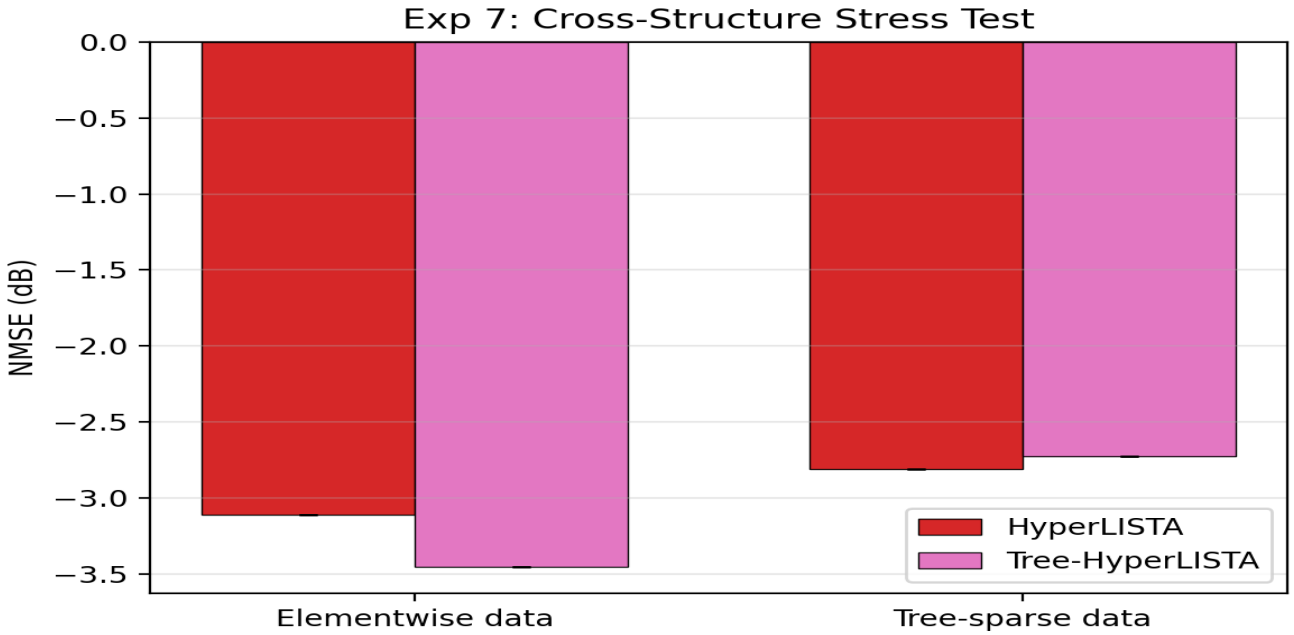
Magnitude: TH robust (–28.7→–28.1 dB at 3×) while HyperLISTA diverges. Consistency: degrades as tree structure breaks.

Appendix G–H: Low-SNR & Superlinear Convergence



TH maintains advantage at all SNR levels. Per-instance tuning → –4.0 dB at only 6 layers (future work direction).

Appendix I: Cross-Structure Experiments



Tree-aware method does NOT hurt performance on non-tree signals (–3.45 dB vs –2.73 dB). Safe to use when structure is uncertain.

Key Takeaways

1. Structure > Parameters: 3 params + tree operator beats 1.56M params by 15.5 dB

2. Minimal Parameterization Generalizes: 3-parameter principle extends to tree-structured sparsity

3. Hard Projection Works: Simple greedy top-K tree projection nearly matches hybrid mode

4. Real-World Validation: Best SSIM on wavelet image CS at all compression ratios

5. Robust & Safe: Graceful degradation under mismatch; doesn't hurt non-tree signals

Thank You

Tree-HyperLISTA: 3 parameters, 12.8 dB better than HyperLISTA, perfect recall

Code: github.com/ALPHAGOD12/tree-hyperlista

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